

Empirical Software Engineering

Write your answers directly on these pages (there's always a risk that loose papers disappear)—use the back also if possible. I'll be at the written exam twice (first time after approximately one hour).

On January 28 at 13.30 you are welcome to Richard's office (4th floor in the Jupiter building at Campus Lindholmen) to complain about the grading. Before the meeting you *must* send Richard an email clearly pointing out where you think the error is, what you wrote, and why you believe the grading was not correct. If I don't receive such an email before 13.30 on January 28, then I will not meet with you.

Statistics is the grammar of science.

— Karl Pearson

Grade 3: 44 points; ~50%

Grade 4: 62 points; ~70%

Grade 5: 79 points; ~90%

Maximum: 88 points

Question 1 :

(8p) Over the years Bayesian data analysis has evolved and spread to all natural sciences. There are several reasons for this, e.g., a principled way to incorporate prior knowledge, increase in computational power making Markov chain Monte Carlo a viable option for sampling, and probabilistic programming languages gaining ground. However, not until lately have statisticians developed guidelines researchers can follow to systematically design Bayesian models.

In *your* opinion, what key steps are compulsory when conducting Bayesian data analysis? Please explain each step one takes when designing models so that we can place *some* confidence in the results.

You can either draw a flowchart and explain each step, or write a numbered list explaining each step.

(It's ok to write on the backside, if they haven't printed on the backside again...)

Solution: There is no standard. However, quite recently Gelman *et al.* (arXiv:2011.01808) published a paper on arXiv explaining what they believed were the key parts in Bayesian data analysis. In short, a lot of the things they talk about in that paper are things we have covered in this course.

In order to get 8 points there need to be a minimum of 4 areas that are listed and explained, e.g., prior or posterior predictive checks, diagnostics, model comparison, etc.

Start with a null model, do prior checks, check diagnostics, do posterior checks, then conduct inferential statistics if needed. Also, it would be good if you mention **comparisons** of models and that it is an **iterative** approach.

Question 2 :

(4p) Underfitting and overfitting are two concepts not always stressed a lot with black-box machine learning approaches. In this course, however, you've probably heard me talk about these concepts a hundred times...

What happens when you underfit and overfit, i.e., **what would the results be?** What are some principled **ways to deal with** under- and overfitting?

Solution: One straightforward way to discuss this is to show two simple models, one where we only have a general intercept α (complete pooling), and one where you estimate an intercept for each cluster in the data (varying intercept), but where we don't share information between the clusters (think of the pond example in the coursebook), i.e., no pooling.

This question can be answered in many different ways.

$$y_i \sim \text{Binomial}(n, p_i)$$

$$\text{logit}(p_i) = \alpha$$

$$\alpha \sim \text{Normal}(0, 2.5)$$

above complete pooling, below no pooling

$$y_i \sim \text{Binomial}(n, p_i)$$

$$\text{logit}(p_i) = \alpha_{\text{CLUSTER}[j]}$$

$$\alpha_j \sim \text{Normal}(0, 2)$$

$$\text{for } j = 1, \dots, n$$

Question 3 :

(10p) To understand **how team size affects psychological safety** the following data was collected:

Team	Team size	SPI	Psychological safety
1	5	67%	High
2	15	33%	Low
3	11	49%	Low
4	7	90%	High
$\vdots$	$\vdots$	$\vdots$	$\vdots$

The experiment started with assuming that planning effectiveness and psychological safety has a very strong association. For planning effectiveness they used schedule performance indicator (SPI) as a stand-in variable.

Schedule Performance Indicator = (Completed points / Planned points)

Based on the result, if the SPI is more than 50% they are classified as a team with high psychological safety. If less than 50% they are classified as a team with low psychological safety.

With the above data, the firm wants to use your knowledge to understand the association between team size and psychological safety.

Write down the mathematical model definition for this prediction using any variable names and priors of your choice.

State the ontological and epistemological reasons for your likelihood. Remember to clearly state and justify the choices and assumptions regarding your model.

Solution: This one is actually a bit trickier than it looks like at first glance. Sure, we know that we should design a model where SPI **or** psychological safety should be used as an outcome, but which one should we use?

SPI is a continuous number from 0 to 100, while psychological safety is a dichotomous variable (basically 1/0). Hence, in the first case a Beta regression could be a possibility, while in the latter a logistic regression could be an option. The lesson here is that we should probably use model comparison to look at out of sample prediction capabilities.

I would be very hesitant in using psychological safety as an outcome. Why? Well, we have taken a continuous value (SPI) and made it into a dichotomous value. In short, from an information theoretical point of view we are shooting ourselves in the foot. Never bin your variables!

To get max point on this question the answer **must** contain a reasoning that we should not use dichotomous variables as a replacement for continuous variables.

Question 4 :

(3p) What is **epistemological justification** and how does it differ from **ontological justification**, when we design models and choose likelihoods? Please provide **an example** where you **argue** epistemological and ontological reasons for selecting a likelihood.

Solution: Epistemological, rooted in information theory and the concept of maximum entropy distributions. Ontological, the way nature works, i.e., a physical assumption about the world.

One example could be the **Normal()** likelihood for real (continuous) numbers where we have additive noise in the process (ontological), and where we know that in such cases **Normal** is the maximum entropy distribution.

Question 5 :

(4p) When diagnosing Markov chains, we often look at several diagnostics to form an opinion of how well things have gone. Name four diagnostics we commonly use? What do we **look for** in each diagnostics (i.e, what thresholds or signs do we look for)? Finally, **what do they tell us**?

Solution:

Effective sample size: A measure of the efficiency of the chains. 10% of total sample size (after warmup) or at least 200 for each parameter.

$\hat{R}$: A measure of the between and the within variance of our chains. Indicates if we have reached a stationary posterior probability distribution. Should approach 1.0 and be below 1.01.

Traceplots: Check if the chains are mixing well. Are they converging around the same parameter space?

MCSE: Markov chain standard error or Standard Error for short.

E-BFMI: etc.

Divergent chains: ...

Question 6 :

(4p) Explain the four main benefits of using multilevel models.

Solution:

- Improved estimates for repeat sampling. When more than one observation arises from the same individual, location, or time, then traditional, single-level models either maximally underfit or overfit the data.
- Improved estimates for imbalance in sampling. When some individuals, locations, or times are sampled more than others, multilevel models automatically cope with differing uncertainty across these clusters. This prevents over-sampled clusters from unfairly dominating inference.
- Estimates of variation. If our research questions include variation among individuals or other groups within the data, then multilevel models are a big help, because they model variation explicitly.
- Avoid averaging, retain variation. Frequently, scholars pre-average some data to construct variables. This can be dangerous, because averaging removes variation, and there are also typically several different ways to perform the averaging. Averaging therefore both manufactures false confidence and introduces arbitrary data transformations. Multilevel models allow us to preserve the uncertainty and avoid data transformations.

Table 1: Output from running WAIC on three models.

	WAIC	SE	dWAIC	dSE	pWAIC
m1	127.6	14.69	0.0	NA	4.7
m3	129.4	15.10	1.8	0.90	5.9
m2	140.6	11.21	13.1	10.82	3.8

Question 7 :

(4p) As a result of comparing three models, we get the above output. What does each column (WAIC, SE, dWAIC, dSE, and pWAIC) **mean**? **Which** model would you **select** based on the output?

Solution:

WAIC: information criteria (the lower the better).

pWAIC: effective number of parameters in the model.

dWAIC: the difference in WAIC between the models.

SE: standard error. How much noise are we not capturing.

dSE: the difference in SE between the models.

If you have 4–6 times larger dWAIC than dSE you have fairly strong indications there is a ‘significant’ difference. Here **no model is, relatively speaking, the best**. If we are forced to pick a model then one could follow to logical arguments: 1. Pick the one with the lowest WAIC. 2. Pick the simplest model.

Question 8 :

(5p) Write an example mathematical model formula for a Poisson regression model with two different kinds of varying intercepts, also known as a cross-classified model.

Solution: See pp. 415 in course book for an example and don't forget to use the sub i notation!

SOLUTION

Question 9 :

(4p) Explain the terms in your own words:

- prior
- posterior
- information entropy
- instrumental variable

Solution: The first three are easiest to explain using the formulas (Bayes' theorem and information entropy). However, one could make a more high-level conceptual explanation.

The final one can be explained in different ways, e.g., in causal terms, an instrumental variable is a variable that acts like a natural experiment on the exposure.

Or in other words, it is a third variable, Z , used in regression analysis when you have endogenous variables—variables that are influenced by other variables in the model, i.e., you use it to account for unexpected behavior between variables. Using an instrumental variable to identify the hidden (unobserved) correlation allows you to see the true correlation between the explanatory variable X and response variable, Y .

Let's say you had two correlated variables that you wanted to regress: X and Y . Their correlation might be described by a third variable Z , which is associated with X in some way. Z is also associated with Y but only through Y 's direct association with X . For example, let's say you wanted to investigate the link between depression (X) and smoking (Y). Lack of job opportunities (Z) could lead to depression, but it is only associated with smoking through its association with depression (i.e. there isn't a direct correlation between lack of job opportunities and smoking). This third variable, Z (lack of job opportunities), can generally be used as an instrumental variable if it can be measured and its behavior can be accounted for.

Question 10 :

(2p) What are the two kinds of varying effects? Explain the effect they have on a statistical model.

Solution: Intercepts and slopes. Our models often make a jump in prediction capability when we add both. Together they are often called varying effects.

A varying intercept means we estimate an intercept for each, e.g., subject in an experiment. Varying slopes allows us to also estimate a separate slope for each subject.

These type of models often result in many parameters that need to be estimated. That is however usually OK, as long as we use partial pooling and an information criteria to keep an eye on overfitting (remember: the more complex models will be penalized).

Question 11 :

(8p) What are the four elemental confounds on which any Directed Acyclic Graph can be explained?

Please draw the four confounds. Explain what they mean (preferably by explaining if one should condition or not on certain elements).

Solution: pp. 185 in the course book.

SOLUTION

Question 12 :

(4p) What is the **purpose** and **limitations** of using laboratory experiments and experimental simulations as a research strategy?

Solution:

Experimental simulations: Real actors' behavior. Non-natural settings (contrived).

Obtrusiveness: High level of obtrusiveness as the context is artificial

Goal: Study a specific setting with higher degree of measurement and context control

Generalizability is reduced as it still mirrors a specific setting

Higher precision of the measurement of the behavior

Lower level of realism

Controlled setting that resembles the real world.

Examples of research methods: Simulation games and Management games

Both qualitative and quantitative (depends on the instrument)

Laboratory experiment: Real actors' behavior. Non-natural settings (contrived). The setting is **more controlled**.

Obtrusiveness: High level of obtrusiveness

Goal: Minimize confounding factors and extraneous conditions. Maximize the precision of the measurement of the behavior. **Establish causality** between variables. Limited number of subjects. Limits the generalizability. Unrealistic context.

Question 13 :

(6p) Below follows an abstract from a research paper. Answer the questions,

- Which of the eight research strategies presented in the ABC framework does this paper likely fit? Justify and argue!
- Can you argue the main validity threats of the paper, based on the research strategy you picked?
 - It would be very good if you can list threats in the four common categories we usually work with in software engineering, i.e., internal, external, construct, and conclusion validity threats.

Context. Computer workers in general, and software developers specifically, are under a high amount of stress due to continuous deadlines and, often, over-commitment.

Objective. This study investigates the effects of a neuroplasticity practice, a specific breathing practice, on the attention awareness, well-being, perceived productivity, and self-efficacy of computer workers.

Method. We created a questionnaire mainly from existing, validated scales as entry and exit survey for data points for comparison before and after the intervention. The intervention was a 12-week program with a weekly live session that included a talk on a well-being topic and a facilitated group breathing session. During the intervention period, we solicited one daily journal note and one weekly well-being rating. We replicated the intervention in a similarly structured 8-week program. The data was analyzed using Bayesian multilevel models for the quantitative part and thematic analysis for the qualitative part.

Results. The intervention showed improvements in participants' experienced inner states despite an ongoing pandemic and intense outer circumstances for most. Over the course of the study, we found an improvement in the participants' ratings of how often they found themselves in good spirits as well as in a calm and relaxed state. We also aggregate a large number of deep inner reflections and growth processes that may not have surfaced for the participants without deliberate engagement in such a program.

Conclusion. The data indicates usefulness and effectiveness of an intervention for computer workers in terms of increasing well-being and resilience. Everyone needs a way to deliberately relax, unplug, and recover. Breathing practice is a simple way to do so, and the results call for establishing a larger body of work to make this common practice.

Solution: I would argue that this should be classified as a Field Experiment.

Question 14 :

(11p) A common research method in software engineering that is used to complement other methods is survey research. Here follows a number of questions connected to survey research:

1. We often differ between reliability and validity concerning surveys,
 - (a) What is the difference between the reliability and validity in survey design? (2p)
 - (b) Name and describe at least two types of reliability in survey design. (3p)
 - (c) Name and describe at least two types of validity in survey design. (3p)
2. Even if you measure and estimate reliability and validity you still want to *evaluate* the survey instrument. Which are the two (2) common ways of evaluating a survey instrument? Explain their differences. (3p)

Solution:

1a. Reliability: A survey is reliable if we administer it many times and get roughly the same distribution of results each time. Validity: How well the survey instrument measures what it is supposed to measure?

1b. e.g., Test-retest (intra-observer), alternate form reliability, internal consistency, inter-observer (Krippendorff's α).

1c. e.g., content validity, criterion validity, construct validity.

2. **Focus groups:** Assemble a group of people representing respondents, experts, beneficiaries of survey results. Ask them to complete the survey questionnaire. The group identifies missing or unnecessary questions, ambiguous questions and instructions. **Pilot studies:** Uses the same procedures as planned for the main study but with smaller sample. Targets not only problems with questions but also response rate and reliability assessment.

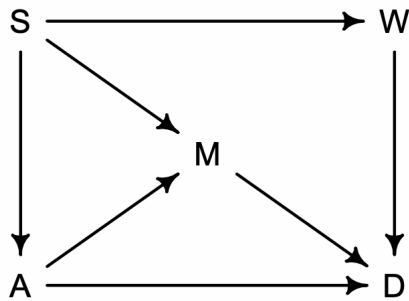


Figure 1: A messy Directed Acyclic Graph.

Question 15 :

(6p) Look at the DAG above. We want to estimate the total causal effect of W on D . Which variable(s) should we condition on and why?

Solution:

```

> library(dagitty)
> dag <- dagitty( "dag {
  A -> D
  A -> M -> D
  A <- S -> M
  S -> W -> D
}" )
> adjustmentSets(dag, exposure="W", outcome="D")

```

and the answer is $\{ A, M \} \setminus \{ S \}$, i.e., either A and M, or only S. In short, check what elementary confounds we have in the DAG (DAGs can only contain four), follow the paths, decide what to condition on.

Question 16 :

(5p or even -5p) You get one point if you answer a question correctly. You get -1 points if you fail to answer the question correctly (if you don't answer you get 0 points). Simply write your answer below.

1. We should always start the Bayesian data analysis by designing a
2. Adding predictors to a model can lead to several things. Two common things are ...
3. My $\hat{R}$ value is 2.6. I should ...
4. We can quantify ... using Kullback-Leibler divergence.
5. I am first and foremost always interested in propagating ... while doing BDA.

Certum est.

Solution:

1. null/initial/simple model
2. overfitting, using a confounder since we might not have a DAG, etc.
3. probably not increase iterations... You must likely reparameterize the model. In short, you're in deep shit and something is seriously wrong.
4. distance to the target, i.e., if we have a pair of candidate distributions, then the candidate that minimizes the divergence will be closest to the target. Since predictive models specify probabilities of events (observations), we can use divergence to compare the accuracy of models.
5. uncertainty